2.34 Integration: Reversing the Chain Rule Part 2

Worked Solutions

1 a)

$\int e^{2x+1}\,dx = \frac{1}{2}e^{2x+1} + c.$

1 b)

$\int e^{5-3x}\,dx = -\frac{1}{3}e^{5-3x} + c.$

1 c)

$\int xe^{x^2}\,dx = x \times \frac{1}{2x}e^{x^2} + c = \frac{1}{2}e^{x^2} + c.$

1 d)

$\int 4x^3e^{6x^4}\,dx = 4x^3 \times \frac{1}{24x^3}e^{6x^4} + c = \frac{1}{6}e^{6x^4} + c.$

2 a)

$\int \cos x\, e^{\sin x}\,dx = \cos x \times \frac{1}{\cos x}e^{\sin x} + c = e^{\sin x} + c.$

2 b)

$\int \sin 3x\, e^{6\cos 3x}\,dx = \sin 3x \times \frac{1}{-18\sin 3x}e^{6\cos 3x} + c = -\frac{1}{18}e^{6\cos 3x} + c.$

2 c)

$\int 4\cos\tfrac{1}{2}x\, e^{\sin\frac{1}{2}x}\,dx = 4\cos\tfrac{1}{2}x \times \frac{1}{\frac{1}{2}\cos\frac{1}{2}x}e^{\sin\frac{1}{2}x} + c = 8e^{\sin\frac{1}{2}x} + c.$

2 d)

$\int 9\sin\tfrac{1}{3}x\, e^{3\cos\frac{1}{3}x}\,dx = 9\sin\tfrac{1}{3}x \times \frac{1}{-\sin\frac{1}{3}x}e^{3\cos\frac{1}{3}x} + c = -9e^{3\cos\frac{1}{3}x} + c.$

3 a)

$\int (\ln 2)2^xe^{2^x}\,dx = (\ln 2)2^x \times \frac{1}{(\ln 2)2^x}e^{2^x} + c = e^{2^x} + c.$

3 b)

$\int 3^xe^{3^x}\,dx = 3^x \times \frac{1}{(\ln 3)3^x}e^{3^x} + c = \frac{1}{\ln 3}e^{3^x} + c.$

3 c)

$\int \frac{1}{\ln 5}5^xe^{5^x+1}\,dx = \frac{1}{\ln 5}5^x \times \frac{1}{(\ln 5)5^x}e^{5^x+1} + c = \frac{1}{(\ln 5)^2}e^{5^x+1} + c.$

3 d)

$\int 4^xe^{4^{x+1}}\,dx = \int 4^xe^{4\times 4^x}\,dx = 4^x \times \frac{1}{4(\ln 4)4^x}e^{4^{x+1}} + c = \frac{1}{4\ln 4}e^{4^{x+1}} + c.$

4 a)

$\int \sec^2 x\, e^{\tan x}\,dx = \sec^2 x \times \frac{1}{\sec^2 x}e^{\tan x} + c = e^{\tan x} + c.$

4 b)

$\int 2\operatorname{cosec} x \cot x\, e^{4\operatorname{cosec} x}\, dx = 2\operatorname{cosec} x \cot x \times \frac{1}{-4\operatorname{cosec} x \cot x} e^{4\operatorname{cosec} x} + c = -\frac{1}{2} e^{4\operatorname{cosec} x} + c.$

4 c)

$\int 3\sec 6x \tan 6x\, e^{\sec 6x}\, dx = 3\sec 6x \tan 6x \times \frac{1}{6\sec 6x \tan 6x} e^{\sec 6x} + c = \frac{1}{2} e^{\sec 6x} + c.$

4 d)

$\int -8\operatorname{cosec}^2 \frac{1}{2}x\, e^{16\cot \frac{1}{2}x}\, dx = -8\operatorname{cosec}^2 \frac{1}{2}x \times \frac{1}{16 \times -\frac{1}{2}\operatorname{cosec}^2 \frac{1}{2}x} e^{16\cot \frac{1}{2}x} + c = e^{16\cot \frac{1}{2}x} + c.$

5)

$\frac{dy}{dx} = -\frac{e^{1/x}}{x^2} \Rightarrow \frac{dy}{dx} = -x^{-2} e^{x^{-1}} \Rightarrow y = -x^{-2} \times \frac{1}{-x^{-2}} e^{x^{-1}} + c \Rightarrow y = e^{1/x} + c.$ When $x = 1$, $y = -e$: $-e = e^{1/1} + c \Rightarrow c = -2e.$ $\therefore y = e^{1/x} - 2e.$

6 a)

$$\begin{aligned}\int_0^1 x^2 e^{x^3} dx &= \left[x^2 \times \frac{1}{3x^2} e^{x^3}\right]_0^1 = \left[\frac{1}{3} e^{x^3}\right]_0^1 \\ &= \left(\frac{1}{3} e^{1^3}\right) - \left(\frac{1}{3} e^{0^3}\right) = \frac{1}{3} e - \frac{1}{3} = \frac{1}{3}(e-1).\end{aligned}$$

6 b)

$$\begin{aligned}\int_0^{\pi/4} \cos 2x\, e^{\sin 2x} dx &= \left[\cos 2x \times \frac{1}{2\cos 2x} e^{\sin 2x}\right]_0^{\pi/4} = \left[\frac{1}{2} e^{\sin 2x}\right]_0^{\pi/4} \\ &= \left(\frac{1}{2} e^{\sin(2\times\pi/4)}\right) - \left(\frac{1}{2} e^{\sin(2\times 0)}\right) = \frac{1}{2} e^1 - \frac{1}{2} e^0 \\ &= \frac{1}{2} e - \frac{1}{2} = \frac{1}{2}(e-1).\end{aligned}$$

6 c)

$$\begin{aligned}\int_1^2 (\ln 8) 2^x e^{2^x} dx &= \left[(3\ln 2) 2^x \times \frac{1}{(\ln 2) 2^x} e^{2^x}\right]_1^2 = \left[3e^{2^x}\right]_1^2 \\ &= 3e^{2^2} - 3e^{2^1} = 3e^4 - 3e^2 = 3e^2(e^2 - 1).\end{aligned}$$

6 d)

$$\begin{aligned}\int_0^{\pi/16} 20\sec^2 4x\, e^{\tan 4x} dx &= \left[20\sec^2 4x \times \frac{1}{4\sec^2 4x} e^{\tan 4x}\right]_0^{\pi/16} = \left[5e^{\tan 4x}\right]_0^{\pi/16} \\ &= 5e^{\tan(4\times\pi/16)} - 5e^{\tan(4\times 0)} = 5e^1 - 5e^0 \\ &= 5e - 5 = 5(e-1).\end{aligned}$$

7)

$$\lim_{\delta x\to 0}\sum_{x=1}^{4}\frac{e^{2\sqrt{x}}}{\sqrt{x}}\delta x=\int_1^4\frac{e^{2\sqrt{x}}}{\sqrt{x}}dx=\int_1^4 x^{-1/2}e^{2x^{1/2}}dx$$
$$=\left[x^{-1/2}\times\frac{1}{x^{-1/2}}e^{2x^{1/2}}\right]_1^4=\left[e^{2x^{1/2}}\right]_1^4$$
$$=e^{2\times 4^{1/2}}-e^{2\times 1^{1/2}}=e^4-e^2=e^2(e^2-1).$$

8 a)

$\int\sin(5x+\frac{\pi}{7})dx=-\frac{1}{5}\cos(5x+\frac{\pi}{7})+c.$

8 b)

$\int 8x\cos 2x^2dx=8x\times\frac{1}{4x}\sin 2x^2+c=2\sin 2x^2+c.$

8 c)

$\int(2x+1)\sin(x^2+x-1)dx=(2x+1)\times-\frac{1}{2x+1}\cos(x^2+x-1)+c=-\cos(x^2+x-1)+c.$

8 d)

$\int(x^2-3)\cos(6x-\frac{2}{3}x^3)dx=(x^2-3)\times\frac{1}{6-2x^2}\sin(6x-\frac{2}{3}x^3)+c=\frac{x^2-3}{-2(x^2-3)}\sin(6x-\frac{2}{3}x^3)+c=-\frac{1}{2}\sin(6x-\frac{2}{3}x^3)+c.$

9 a)

$\int\cos x\sin(\sin x)dx=\cos x\times-\frac{1}{\cos x}\cos(\sin x)+c=-\cos(\sin x)+c.$

9 b)

$\int\sin x\sin(\cos x)dx=\sin x\times-\frac{1}{-\sin x}\cos(\cos x)+c=\cos(\cos x)+c.$

9 c)

$\int\cos x\cos(\sin x)dx=\cos x\times\frac{1}{\cos x}\sin(\sin x)+c=\sin(\sin x)+c.$

9 d)

$\int\sin x\cos(\cos x)dx=\sin x\times\frac{1}{-\sin x}\sin(\cos x)+c=-\sin(\cos x)+c.$

10 a)

$\int(\ln 5)5^x\sin(5^x)dx=(\ln 5)5^x\times-\frac{1}{(\ln 5)5^x}\cos(5^x)+c=-\cos(5^x)+c.$

10 b)

$\int 9^x\cos(9^x)dx=9^x\times\frac{1}{(\ln 9)9^x}\sin(9^x)+c=\frac{1}{\ln 9}\sin(9^x)+c.$

10 c)

$\int\frac{\sin(\ln x)}{x}dx=\int\frac{1}{x}\sin(\ln x)dx=\frac{1}{x}\times-\frac{1}{1/x}\cos(\ln x)+c=-\cos(\ln x)+c.$

10 d)

$\int \frac{\cos(2\ln x)}{x}dx = \int \frac{1}{x}\cos(2\ln x)dx = \frac{1}{x} \times \frac{1}{2/x}\sin(2\ln x) + c = \frac{1}{x} \times \frac{x}{2}\sin(2\ln x) + c = \frac{1}{2}\sin(2\ln x) + c.$

11 a)

$\frac{dy}{dx} = \operatorname{cosec}^2 2x \sin(\cot 2x) \Rightarrow y = \operatorname{cosec}^2 2x \times \frac{1}{-2\operatorname{cosec}^2 2x} \times -\cos(\cot 2x) + c \Rightarrow y = \frac{1}{2}\cos(\cot 2x) + c.$

11 b)

$\frac{dy}{dx} = 4\operatorname{cosec} 3x \cot 3x \cos(8\operatorname{cosec} 3x) \Rightarrow y = 4\operatorname{cosec} 3x \cot 3x \times \frac{1}{-24\operatorname{cosec} 3x \cot 3x}\sin(8\operatorname{cosec} 3x) + c \Rightarrow$
$y = -\frac{1}{6}\sin(8\operatorname{cosec} 3x) + c.$

11 c)

$\frac{dy}{dx} = 9\sec^2 \frac{1}{3}x \sin(90\tan \frac{1}{3}x)$
$\Rightarrow y = 9\sec^2 \frac{1}{3}x \times -\frac{1}{30\sec^2 \frac{1}{3}x}\cos(90\tan \frac{1}{3}x) + c$
$\Rightarrow y = -\frac{3}{10}\cos(90\tan \frac{1}{3}x) + c.$

11 d)

$\frac{dy}{dx} = \sec \frac{2}{3}x \tan \frac{2}{3}x \cos(18\sec \frac{2}{3}x) \Rightarrow y = \sec \frac{2}{3}x \tan \frac{2}{3}x \times \frac{1}{12\sec \frac{2}{3}x \tan \frac{2}{3}x}\sin(18\sec \frac{2}{3}x) + c \Rightarrow y = \frac{1}{12}\sin(18\sec \frac{2}{3}x) + c.$

12)

$\frac{dy}{dx} = 12x\sin x^2 \Rightarrow y = 12x \times -\frac{1}{2x}\cos x^2 + c \Rightarrow y = -6\cos x^2 + c.$ When $x = \frac{\sqrt{\pi}}{2}$, $y = 24$:
$24 = -6\cos(\frac{\sqrt{\pi}}{2})^2 + c \Rightarrow 24 = -6\cos\frac{\pi}{4} + c \Rightarrow c = 24 + 3\sqrt{2}.$ $\therefore y = -6\cos x^2 + 24 + 3\sqrt{2}.$

13 a)

$$\begin{aligned}\int_{\ln(\pi/2)}^{\ln\pi} e^x \sin(e^x)dx &= \left[e^x \times -\frac{1}{e^x}\cos(e^x)\right]_{\ln(\pi/2)}^{\ln\pi} = [-\cos(e^x)]_{\ln(\pi/2)}^{\ln\pi} \\ &= (-\cos(e^{\ln\pi})) - (-\cos(e^{\ln(\pi/2)})) \\ &= (-\cos\pi) - (-\cos\tfrac{\pi}{2}) = 1 - 0 = 1.\end{aligned}$$

13 b)

$$\begin{aligned}\int_0^{\sqrt{\pi}/4} 40x\cos 8x^2 dx &= \left[40x \times \frac{1}{16x}\sin 8x^2\right]_0^{\sqrt{\pi}/4} = \left[\frac{5}{2}\sin 8x^2\right]_0^{\sqrt{\pi}/4} \\ &= \frac{5}{2}\sin\left(8(\tfrac{\sqrt{\pi}}{4})^2\right) - \frac{5}{2}\sin(8(0)^2) \\ &= \frac{5}{2}\sin\frac{\pi}{2} - \frac{5}{2}\sin 0 = \frac{5}{2} \times 1 - \frac{5}{2} \times 0 = \frac{5}{2}.\end{aligned}$$

13 c)

$$\begin{aligned}\int_{\sqrt{2\pi}}^{\sqrt{3\pi}} 18x\sin x^2 dx &= \left[18x \times -\frac{1}{2x}\cos x^2\right]_{\sqrt{2\pi}}^{\sqrt{3\pi}} = [-9\cos x^2]_{\sqrt{2\pi}}^{\sqrt{3\pi}} \\ &= (-9\cos(\sqrt{3\pi})^2) - (-9\cos(\sqrt{2\pi})^2) \\ &= (-9\cos 3\pi) - (-9\cos 2\pi) = (-9\times -1) - (-9\times 1) = 9-(-9) = 18.\end{aligned}$$

13 d)

$$\begin{aligned}\int_{\pi/4}^{\pi/3}(24x-7\pi)\cos(12x^2-7\pi x+\pi^2)dx &= \left[(24x-7\pi)\times\frac{1}{24x-7\pi}\sin(12x^2-7\pi x+\pi^2)\right]_{\pi/4}^{\pi/3} \\ &= [\sin(12x^2-7\pi x+\pi^2)]_{\pi/4}^{\pi/3} \\ &= \sin\left(12(\tfrac{\pi}{3})^2-7\pi(\tfrac{\pi}{3})+\pi^2\right) \\ &\quad -\sin\left(12(\tfrac{\pi}{4})^2-7\pi(\tfrac{\pi}{4})+\pi^2\right) \\ &= \sin\left(\frac{4}{3}\pi^2-\frac{7}{3}\pi^2+\pi^2\right)-\sin\left(\frac{3}{4}\pi^2-\frac{7}{4}\pi^2+\pi^2\right) \\ &= \sin 0-\sin 0 = 0.\end{aligned}$$

14)

$$\begin{aligned}\lim_{\delta x\to 0}\sum_{x=16/\pi}^{8/\pi}\frac{\sin(4/x)}{8x^2}\delta x &= \int_{16/\pi}^{8/\pi}\frac{\sin(4/x)}{8x^2}dx = \int_{16/\pi}^{8/\pi}\frac{1}{8}x^{-2}\sin(4x^{-1})dx \\ &= \left[\frac{1}{8}x^{-2}\times -\frac{1}{-4x^{-2}}\cos(4x^{-1})\right]_{16/\pi}^{8/\pi} = \left[\frac{1}{32}\cos(\tfrac{4}{x})\right]_{16/\pi}^{8/\pi} \\ &= \frac{1}{32}\cos\left(\frac{4}{8/\pi}\right)-\frac{1}{32}\cos\left(\frac{4}{16/\pi}\right) \\ &= \frac{1}{32}\cos\frac{\pi}{2}-\frac{1}{32}\cos\frac{\pi}{4} = \frac{1}{32}\times 0-\frac{1}{32}\times\frac{\sqrt{2}}{2} \\ &= 0-\frac{\sqrt{2}}{64} = -\frac{\sqrt{2}}{64}.\end{aligned}$$

15 a)

$\int \frac{1}{x+1}dx = \frac{1}{1}\times\ln|x+1|+c = \ln|x+1|+c.$

15 b)

$\int \frac{2}{2x-5}dx = \frac{2}{2}\times\ln|2x-5|+c = \ln|2x-5|+c.$

15 c)

$\int \frac{8}{4x+3}dx = \frac{8}{4}\times\ln|4x+3|+c = 2\ln|4x+3|+c.$

15 d)

$\int \frac{9}{5x-7}dx = \frac{9}{5} \times \ln|5x-7| + c = \frac{9}{5}\ln|5x-7| + c.$

15 e)

$\int \frac{3}{1-4x}dx = \frac{3}{-4} \times \ln|1-4x| + c = -\frac{3}{4}\ln|1-4x| + c.$

15 f)

$\int \frac{15}{5-3x}dx = \frac{15}{-3} \times \ln|5-3x| + c = -5\ln|5-3x| + c.$

16 a)

$\int \frac{2x}{x^2+3}dx = \frac{2x}{2x} \times \ln|x^2+3| + c = \ln|x^2+3| + c.$

16 b)

$\int \frac{3x^2}{5+x^3}dx = \frac{3x^2}{3x^2} \times \ln|5+x^3| + c = \ln|5+x^3| + c.$

16 c)

$\int \frac{15x}{5x^2-4}dx = \frac{15x}{10x} \times \ln|5x^2-4| + c = \frac{3}{2}\ln|5x^2-4| + c.$

16 d)

$\int \frac{10x^3}{6-x^4}dx = \frac{10x^3}{-4x^3} \times \ln|6-x^4| + c = -\frac{5}{2}\ln|6-x^4| + c.$

17 a)

$\int \frac{2x+4}{x^2+4x-3}dx = \frac{2x+4}{2x+4} \times \ln|x^2+4x-3| + c = \ln|x^2+4x-3| + c.$

17 b)

$\int \frac{3x^2-3}{x^3-3x+9}dx = \frac{3x^2-3}{3x^2-3} \times \ln|x^3-3x+9| + c = \ln|x^3-3x+9| + c.$

17 c)

$\int \frac{x+1}{4x^2+8x+5}dx = \frac{x+1}{8x+8} \times \ln|4x^2+8x+5| + c = \frac{x+1}{8(x+1)} \times \ln|4x^2+8x+5| + c = \frac{1}{8}\ln|4x^2+8x+5| + c.$

17 d)

$\int \frac{6x^2-24x}{x^3-6x^2-1}dx = \frac{6x^2-24x}{3x^2-12x} \times \ln|x^3-6x^2-1| + c = \frac{2(3x^2-12x)}{3x^2-12x} \times \ln|x^3-6x^2-1| + c = 2\ln|x^3-6x^2-1| + c.$

18 a)

$\int \frac{e^x}{e^x+1}dx = \frac{e^x}{e^x} \times \ln|e^x+1| + c = \ln|e^x+1| + c.$

18 b)

$\int \frac{(\ln 2)2^x-1}{2^x-x}dx = \frac{(\ln 2)2^x-1}{(\ln 2)2^x-1} \times \ln|2^x-x| + c = \ln|2^x-x| + c.$

18 c)

$\int \tan x\,dx = \int \frac{\sin x}{\cos x}dx = \frac{\sin x}{-\sin x} \times \ln|\cos x| + c = -\ln|\cos x| + c = \ln|(\cos x)^{-1}| + c = \ln|\sec x| + c.$

18 d)

$\int \cot x\,dx = \int \frac{\cos x}{\sin x}dx = \frac{\cos x}{\cos x} \times \ln|\sin x| + c = \ln|\sin x| + c.$

19 a)

$\int \frac{\cos 2x}{\sin 2x+1}dx = \frac{\cos 2x}{2\cos 2x} \times \ln|\sin 2x + 1| + c = \frac{1}{2}\ln|\sin 2x + 1| + c.$

19 b)

$\int \frac{6\sin 3x}{4-\cos 3x}dx = \frac{6\sin 3x}{3\sin 3x} \times \ln|4 - \cos 3x| + c = 2\ln|4 - \cos 3x| + c.$

19 c)

$\int \frac{24\cos 6x}{3\sin 6x-5}dx = \frac{24\cos 6x}{18\cos 6x} \times \ln|3\sin 6x - 5| + c = \frac{4}{3}\ln|3\sin 6x - 5| + c.$

19 d)

$\int \frac{18\sin 9x}{15\cos 9x+4}dx = \frac{18\sin 9x}{-135\sin 9x} \times \ln|15\cos 9x + 4| + c = -\frac{2}{15}\ln|15\cos 9x + 4| + c.$

20 a)

$\frac{dy}{dx} = \frac{(\ln 3)3^x + e^x}{3^x + e^x} \Rightarrow y = \frac{(\ln 3)3^x + e^x}{(\ln 3)3^x + e^x} \times \ln|3^x + e^x| + c \Rightarrow y = \ln|3^x + e^x| + c.$

20 b)

$\frac{dy}{dx} = \frac{8 + 4\sec^2 2x}{4x + \tan 2x} \Rightarrow y = \frac{8 + 4\sec^2 2x}{4 + 2\sec^2 2x} \times \ln|4x + \tan 2x| + c = \frac{2(4 + 2\sec^2 2x)}{4 + 2\sec^2 2x} \times \ln|4x + \tan 2x| + c \Rightarrow$
$y = 2\ln|4x + \tan 2x| + c.$

20 c)

$\frac{dy}{dx} = \frac{\cos x + \frac{1}{2\sqrt{x}}}{\sin x + \sqrt{x}} \Rightarrow \frac{dy}{dx} = \frac{\cos x + \frac{1}{2}x^{-1/2}}{\sin x + x^{1/2}} \Rightarrow y = \frac{\cos x + \frac{1}{2}x^{-1/2}}{\cos x + \frac{1}{2}x^{-1/2}} \times \ln|\sin x + x^{1/2}| + c \Rightarrow y =$
$\ln|\sin x + x^{1/2}| + c.$

20 d)

$\frac{dy}{dx} = \frac{30e^{3x} - 20e^{2x}}{e^{2x} - e^{3x}} \Rightarrow y = \frac{30e^{3x} - 20e^{2x}}{2e^{2x} - 3e^{3x}} \times \ln|e^{2x} - e^{3x}| + c = \frac{-10(2e^{2x} - 3e^{3x})}{2e^{2x} - 3e^{3x}} \times \ln|e^{2x} - e^{3x}| + c \Rightarrow$
$y = -10\ln|e^{2x} - e^{3x}| + c.$

21)

$\frac{dy}{dx} = \frac{2x+1}{(x-1)(x+2)} \Rightarrow \frac{dy}{dx} = \frac{2x+1}{x^2+x-2} \Rightarrow y = \frac{2x+1}{2x+1} \times \ln|x^2 + x - 2| + c \Rightarrow y = \ln|x^2 + x - 2| + c.$
When $x = 2$, $y = 2\ln 2$: $2\ln 2 = \ln|2^2 + 2 - 2| + c \Rightarrow 2\ln 2 = \ln|4| + c \Rightarrow 2\ln 2 = 2\ln 2 + c \Rightarrow c = 0.$
$\therefore y = \ln|x^2 + x - 2|.$

22 a)

$$\begin{aligned}\int_0^1 \frac{12}{3x+1}dx &= \left[\frac{12}{3} \times \ln|3x+1|\right]_0^1 = [4\ln|3x+1|]_0^1 \\ &= (4\ln|3\times 1+1|) - (4\ln|3\times 0+1|) \\ &= 4\ln 4 - 4\ln 1 = 4\ln 4 - 4\times 0 = 4\ln 4.\end{aligned}$$

22 b)

$$\begin{aligned}\int_{\sqrt{2}}^{\sqrt{3}} \frac{3x}{x^2-1}dx &= \left[\frac{3x}{2x} \times \ln|x^2-1|\right]_{\sqrt{2}}^{\sqrt{3}} = \left[\frac{3}{2}\ln|x^2-1|\right]_{\sqrt{2}}^{\sqrt{3}} \\ &= \frac{3}{2}\ln|(\sqrt{3})^2-1| - \frac{3}{2}\ln|(\sqrt{2})^2-1| \\ &= \frac{3}{2}\ln 2 - \frac{3}{2}\ln 1 = \frac{3}{2}\ln 2 - \frac{3}{2}\times 0 = \frac{3}{2}\ln 2.\end{aligned}$$

22 c)

$$\begin{aligned}\int_{\pi}^{2\pi} \frac{\sin x}{\cos x+2}dx &= \left[\frac{\sin x}{-\sin x} \times \ln|\cos x+2|\right]_{\pi}^{2\pi} = [-\ln|\cos x+2|]_{\pi}^{2\pi} \\ &= (-\ln|\cos 2\pi+2|) - (-\ln|\cos\pi+2|) \\ &= (-\ln|1+2|) - (-\ln|-1+2|) = (-\ln 3) - (-\ln 1) \\ &= (-\ln 3) - (-0) = -\ln 3 = \ln 3^{-1} = \ln\frac{1}{3}.\end{aligned}$$

22 d)

$$\begin{aligned}\int_{\ln 2}^{\ln 4} \frac{4e^x}{4+e^x}dx &= \left[\frac{4e^x}{e^x} \times \ln|4+e^x|\right]_{\ln 2}^{\ln 4} = [4\ln|4+e^x|]_{\ln 2}^{\ln 4} \\ &= (4\ln|4+e^{\ln 4}|) - (4\ln|4+e^{\ln 2}|) \\ &= 4\ln|4+4| - 4\ln|4+2| = 4\ln 8 - 4\ln 6 \\ &= 4(\ln 8 - \ln 6) = 4\ln\frac{8}{6} = 4\ln\frac{4}{3}.\end{aligned}$$

23)

$$\begin{aligned}\lim_{\delta x\to 0}\sum_{x=1}^{3}\frac{6x+1}{6x^2+2x+1}\delta x = \int_1^3 \frac{6x+1}{6x^2+2x+1}dx &= \left[\frac{6x+1}{12x+2} \times \ln|6x^2+2x+1|\right]_1^3 \\ &= \left[\frac{6x+1}{2(6x+1)} \times \ln|6x^2+2x+1|\right]_1^3 = \left[\frac{1}{2}\ln|6x^2+2x+1|\right]_1^3 \\ &= \frac{1}{2}\ln|6\times 3^2+2\times 3+1| - \frac{1}{2}\ln|6\times 1^2+2\times 1+1| \\ &= \frac{1}{2}\ln 61 - \frac{1}{2}\ln 9 = \frac{1}{2}(\ln 61 - \ln 9) = \frac{1}{2}\ln\frac{61}{9}.\end{aligned}$$

24 a)

$\int e^x(e^x+5)^3dx = e^x \times \frac{1}{e^x} \times \frac{1}{4}(e^x+5)^4 + c = \frac{1}{4}(e^x+5)^4 + c.$

24 b)

$\int e^{3x}(e^{3x}-3)^4dx = e^{3x} \times \frac{1}{3e^{3x}} \times \frac{1}{5}(e^{3x}-3)^5 + c = \frac{1}{15}(e^{3x}-3)^5 + c.$

24 c)

$\int 2e^{4x}(4e^{4x}-1)^5dx = 2e^{4x} \times \frac{1}{16e^{4x}} \times \frac{1}{6}(4e^{4x}-1)^6 + c = \frac{1}{48}(4e^{4x}-1)^6 + c.$

24 d)

$\int 10e^{x/2}\sqrt{e^{x/2}+2}\,dx = \int 10e^{x/2}(e^{x/2}+2)^{1/2}dx = 10e^{x/2} \times \frac{1}{\frac{1}{2}e^{x/2}} \times \frac{2}{3}(e^{x/2}+2)^{3/2} + c = \frac{40}{3}(e^{x/2}+2)^{3/2} + c.$

25 a)

$\int \cos x \sin^2 x\,dx = \int \cos x(\sin x)^2dx = \cos x \times \frac{1}{\cos x} \times \frac{1}{3}(\sin x)^3 + c = \frac{1}{3}\sin^3 x + c.$

25 b)

$\int \cos x \sin^3 x\,dx = \int \cos x(\sin x)^3dx = \cos x \times \frac{1}{\cos x} \times \frac{1}{4}(\sin x)^4 + c = \frac{1}{4}\sin^4 x + c.$

25 c)

$\int \sin x \cos^4 x\,dx = \int \sin x(\cos x)^4dx = \sin x \times \frac{1}{-\sin x} \times \frac{1}{5}(\cos x)^5 + c = -\frac{1}{5}\cos^5 x + c.$

25 d)

$\int \sin 2x \cos^5 2x\,dx = \sin 2x \times \frac{1}{-2\sin 2x} \times \frac{1}{6}(\cos 2x)^6 + c = -\frac{1}{12}\cos^6 2x + c.$

26 a)

$\int \sec^2 x(\tan x - 4)^5dx = \sec^2 x \times \frac{1}{\sec^2 x} \times \frac{1}{6}(\tan x - 4)^6 + c = \frac{1}{6}(\tan x - 4)^6 + c.$

26 b)

$\int \sec^2 3x \tan^3 3x\,dx = \int \sec^2 3x(\tan 3x)^3dx = \sec^2 3x \times \frac{1}{3\sec^2 3x} \times \frac{1}{4}(\tan 3x)^4 + c = \frac{1}{12}\tan^4 3x + c.$

26 c)

$\int 2^x(2^x+4)^4dx = 2^x \times \frac{1}{(\ln 2)2^x} \times \frac{1}{5}(2^x+4)^5 + c = \frac{1}{5\ln 2}(2^x+4)^5 + c.$

26 d)

$\int 3^{x+1}(3^x-2)^5dx = \int 3 \times 3^x(3^x-2)^5dx = 3 \times 3^x \times \frac{1}{(\ln 3)3^x} \times \frac{1}{6}(3^x-2)^6 + c = \frac{1}{2\ln 3}(3^x-2)^6 + c.$

27 a)

$\frac{dy}{dx} = \frac{(\ln x+3)^3}{x} \Rightarrow \frac{dy}{dx} = \frac{1}{x} \times (\ln x+3)^3 \Rightarrow y = \frac{1}{x} \times \frac{1}{1/x} \times \frac{1}{4}(\ln x+3)^4 + c \Rightarrow y = \frac{1}{4}(\ln x+3)^4 + c.$

27 b)

$\frac{dy}{dx} = \frac{(2\ln x+5)^5}{12x} \Rightarrow \frac{dy}{dx} = \frac{1}{12x} \times (2\ln x+5)^5 \Rightarrow y = \frac{1}{12x} \times \frac{x}{2} \times \frac{1}{6}(2\ln x+5)^6 + c \Rightarrow y = \frac{1}{144}(2\ln x+5)^6 + c.$

27 c)

$\frac{dy}{dx} = \sec x \tan x(\sec x + 2)^3 \Rightarrow y = \sec x \tan x \times \frac{1}{\sec x \tan x} \times \frac{1}{4}(\sec x + 2)^4 + c \Rightarrow y = \frac{1}{4}(\sec x + 2)^4 + c.$

27 d)

$\frac{dy}{dx} = \operatorname{cosec}^8 x \cot x \Rightarrow \frac{dy}{dx} = \operatorname{cosec} x \cot x (\operatorname{cosec} x)^7 \Rightarrow y = \operatorname{cosec} x \cot x \times \frac{1}{-\operatorname{cosec} x \cot x} \times \frac{1}{8}(\operatorname{cosec} x)^8 + c \Rightarrow y = -\frac{1}{8}\operatorname{cosec}^8 x + c.$

28 a)

$$\begin{aligned}\int_{\ln 2}^{\ln 4} e^x(4e^x - 5)^3 dx &= \left[e^x \times \frac{1}{4e^x} \times \frac{1}{4}(4e^x - 5)^4\right]_{\ln 2}^{\ln 4} = \left[\frac{1}{16}(4e^x - 5)^4\right]_{\ln 2}^{\ln 4} \\ &= \frac{1}{16}(4e^{\ln 4} - 5)^4 - \frac{1}{16}(4e^{\ln 2} - 5)^4 \\ &= \frac{1}{16}(4 \times 4 - 5)^4 - \frac{1}{16}(4 \times 2 - 5)^4 \\ &= \frac{14641}{16} - \frac{81}{16} = 910.\end{aligned}$$

28 b)

$$\begin{aligned}\int_0^{\pi/2} \cos x \sin^4 x \, dx &= \int_0^{\pi/2} \cos x (\sin x)^4 dx \\ &= \left[\cos x \times \frac{1}{\cos x} \times \frac{1}{5}(\sin x)^5\right]_0^{\pi/2} = \left[\frac{1}{5}(\sin x)^5\right]_0^{\pi/2} \\ &= \frac{1}{5}(\sin \tfrac{\pi}{2})^5 - \frac{1}{5}(\sin 0)^5 = \frac{1}{5} \times 1 - \frac{1}{5} \times 0 = \frac{1}{5}.\end{aligned}$$

28 c)

$$\begin{aligned}\int_{2\pi}^{3\pi} \sin \tfrac{1}{2}x \cos^5 \tfrac{1}{2}x \, dx &= \int_{2\pi}^{3\pi} \sin \tfrac{1}{2}x (\cos \tfrac{1}{2}x)^5 dx \\ &= \left[\sin \tfrac{1}{2}x \times \frac{1}{-\frac{1}{2}\sin \frac{1}{2}x} \times \frac{1}{6}(\cos \tfrac{1}{2}x)^6\right]_{2\pi}^{3\pi} = \left[-\frac{1}{3}(\cos \tfrac{1}{2}x)^6\right]_{2\pi}^{3\pi} \\ &= -\frac{1}{3}\left(\cos(\tfrac{1}{2} \times 3\pi)\right)^6 - \left(-\frac{1}{3}\left(\cos(\tfrac{1}{2} \times 2\pi)\right)^6\right) \\ &= -\frac{1}{3}(\cos \tfrac{3\pi}{2})^6 - \left(-\frac{1}{3}(\cos \pi)^6\right) \\ &= -\frac{1}{3} \times 0 - \left(-\frac{1}{3} \times 1\right) = 0 - (-\tfrac{1}{3}) = \frac{1}{3}.\end{aligned}$$

28 d)

$$\int_e^{e^2} \frac{(\ln x + 3)^5}{2x} dx = \int_e^{e^2} \frac{1}{2x} \times (\ln x + 3)^5 dx$$

$$= \left[\frac{1}{2x} \times \frac{1}{1/x} \times \frac{1}{6}(\ln x + 3)^6\right]_e^{e^2} = \left[\frac{1}{12}(\ln x + 3)^6\right]_e^{e^2}$$
$$= \frac{1}{12}(\ln e^2 + 3)^6 - \frac{1}{12}(\ln e + 3)^6$$
$$= \frac{1}{12}(2+3)^6 - \frac{1}{12}(1+3)^6 = \frac{15625}{12} - \frac{1024}{3} = \frac{3843}{4}.$$

29)

$$\int_k^{2k} \frac{8}{4x+k}dx = \left[\frac{8}{4} \times \ln|4x+k|\right]_k^{2k} = [2\ln|4x+k|]_k^{2k}$$
$$= 2\ln|4\times 2k + k| - 2\ln|4k+k| = 2\ln 9k - 2\ln 5k$$
$$= 2(\ln 9k - \ln 5k) = 2\ln\frac{9k}{5k} = 2\ln\frac{9}{5}.$$

30)

$$\int_{1/k}^{2/k} \frac{k}{kx+5}dx = \left[\frac{k}{k} \times \ln|kx+5|\right]_{1/k}^{2/k} = [\ln|kx+5|]_{1/k}^{2/k}$$
$$= \ln|k \times \tfrac{2}{k} + 5| - \ln|k \times \tfrac{1}{k} + 5|$$
$$= \ln|2+5| - \ln|1+5| = \ln 7 - \ln 6 = \ln\frac{7}{6}.$$

31)

$$\int_{2/k}^{3/k} \frac{k^2x}{k^2x^2-2}dx = \left[\frac{k^2x}{2k^2x} \times \ln|k^2x^2-2|\right]_{2/k}^{3/k} = \left[\frac{1}{2}\ln|k^2x^2-2|\right]_{2/k}^{3/k}$$
$$= \frac{1}{2}\ln|k^2(\tfrac{3}{k})^2 - 2| - \frac{1}{2}\ln|k^2(\tfrac{2}{k})^2 - 2|$$
$$= \frac{1}{2}\ln|9-2| - \frac{1}{2}\ln|4-2| = \frac{1}{2}\ln 7 - \frac{1}{2}\ln 2$$
$$= \frac{1}{2}(\ln 7 - \ln 2) = \frac{1}{2}\ln\frac{7}{2}.$$

32)

$$\int_{\sqrt{k}}^{\sqrt{4k}} 8xe^{x^2}dx = \left[8x \times \frac{1}{2x} \times e^{x^2}\right]_{\sqrt{k}}^{\sqrt{4k}} = [4e^{x^2}]_{\sqrt{k}}^{\sqrt{4k}}$$
$$= 4e^{(\sqrt{4k})^2} - 4e^{(\sqrt{k})^2} = 4e^{4k} - 4e^k = 4e^k(e^{3k}-1).$$

33)

$$\begin{aligned}\int_{\ln(\sqrt{\pi}/4)}^{\ln(\sqrt{\pi}/2)} 8e^{2x}\sin(4e^{2x})dx &= \left[8e^{2x}\times\frac{1}{8e^{2x}}\times-\cos(4e^{2x})\right]_{\ln(\sqrt{\pi}/4)}^{\ln(\sqrt{\pi}/2)}\\ &= [-\cos(4e^{2x})]_{\ln(\sqrt{\pi}/4)}^{\ln(\sqrt{\pi}/2)}\\ &= -\cos(4e^{2\ln(\sqrt{\pi}/2)}) - (-\cos(4e^{2\ln(\sqrt{\pi}/4)}))\\ &= -\cos(4e^{\ln(\pi/4)}) - (-\cos(4e^{\ln(\pi/16)}))\\ &= -\cos(4\times\tfrac{\pi}{4}) - (-\cos(4\times\tfrac{\pi}{16}))\\ &= (-\cos\pi) - (-\cos\tfrac{\pi}{4}) = 1-(-\tfrac{\sqrt{2}}{2}) = 1+\frac{\sqrt{2}}{2}.\end{aligned}$$

34)

$$\begin{aligned}\int_0^2 \frac{2x^2}{4x^3+1}dx &= \left[2x^2\times\frac{1}{12x^2}\times\ln|4x^3+1|\right]_0^2 = \left[\frac{1}{6}\ln|4x^3+1|\right]_0^2\\ &= \frac{1}{6}\ln|4\times2^3+1| - \frac{1}{6}\ln|4\times0^3+1|\\ &= \frac{1}{6}\ln 33 - \frac{1}{6}\ln 1 = \frac{1}{6}\ln 33 - \frac{1}{6}\times 0 = \frac{1}{6}\ln 33.\end{aligned}$$

35)

$$\begin{aligned}\int_0^1 \frac{3}{3x+1}dx - \int_0^1 \frac{3}{(3x+1)^2}dx &= \int_0^1 \left(\frac{3}{3x+1} - \frac{3}{(3x+1)^2}\right)dx\\ &= \int_0^1 \left(\frac{3}{3x+1} - 3(3x+1)^{-2}\right)dx\\ &= \left[3\times\frac{1}{3}\ln|3x+1| - 3\times\frac{1}{3}\times\frac{1}{-1}(3x+1)^{-1}\right]_0^1\\ &= \left[\ln|3x+1| + \frac{1}{3x+1}\right]_0^1\\ &= \left(\ln|3\times1+1| + \frac{1}{3\times1+1}\right) - \left(\ln|3\times0+1| + \frac{1}{3\times0+1}\right)\\ &= (\ln 4 + \tfrac{1}{4}) - (\ln 1 + 1) = \ln 4 - \frac{3}{4}.\end{aligned}$$

36)

$$\begin{aligned}\int_0^{\pi/12} \sin 2x\cos^5 2x\,dx &= \int_0^{\pi/12} \sin 2x(\cos 2x)^5 dx\\ &= \left[\sin 2x\times\frac{1}{-2\sin 2x}\times\frac{1}{6}(\cos 2x)^6\right]_0^{\pi/12} = \left[-\frac{1}{12}(\cos 2x)^6\right]_0^{\pi/12}\\ &= -\frac{1}{12}\left(\cos(2\times\tfrac{\pi}{12})\right)^6 - \left(-\frac{1}{12}(\cos(2\times0))^6\right)\\ &= -\frac{1}{12}(\cos\tfrac{\pi}{6})^6 - \left(-\frac{1}{12}(\cos 0)^6\right)\end{aligned}$$

$$= -\frac{1}{12}(\tfrac{\sqrt{3}}{2})^6 - \left(-\frac{1}{12}(1)^6\right)$$
$$= -\frac{1}{12} \times \frac{27}{64} - (-\tfrac{1}{12}) = -\frac{9}{256} - (-\tfrac{1}{12}) = \frac{37}{768}.$$

When $x = \frac{\pi}{12}$, $y = \sin(2 \times \frac{\pi}{12}) \cos^5(2 \times \frac{\pi}{12}) = \sin\frac{\pi}{6} \cos^5\frac{\pi}{6} = \frac{1}{2} \times (\frac{\sqrt{3}}{2})^5 = \frac{9\sqrt{3}}{64}$. $\therefore$ Area $= \frac{37}{768} - \frac{1}{2} \times \frac{\pi}{12} \times \frac{9\sqrt{3}}{64} = \frac{37}{768} - \frac{3\sqrt{3}\pi}{512}$.

37)

$$y = \frac{\sin 3x}{\cos 3x + 2} \Rightarrow \frac{dy}{dx} = \frac{(\cos 3x + 2) \times 3\cos 3x - \sin 3x(-3\sin 3x)}{(\cos 3x + 2)^2}$$
$$= \frac{3\cos^2 3x + 6\cos 3x + 3\sin^2 3x}{(\cos 3x + 2)^2}$$
$$= \frac{3(\cos^2 3x + \sin^2 3x) + 6\cos 3x}{(\cos 3x + 2)^2} = \frac{3 + 6\cos 3x}{(\cos 3x + 2)^2}.$$

Turning points exist when $\frac{dy}{dx} = 0$: $\dfrac{3 + 6\cos 3x}{(\cos 3x + 2)^2} = 0 \Rightarrow 3 + 6\cos 3x = 0 \Rightarrow \cos 3x = -\frac{1}{2} \Rightarrow 3x = \arccos(-\frac{1}{2}) = \frac{2\pi}{3}$. $3x = \frac{2\pi}{3} \Rightarrow x = \frac{2\pi}{9}$. Or $3x = -\frac{2\pi}{3} \Rightarrow x = -\frac{2\pi}{9}$.

$$\text{When } x = \frac{2\pi}{9}, \quad y = \frac{\sin(3 \times \frac{2\pi}{9})}{\cos(3 \times \frac{2\pi}{9}) + 2} = \frac{\sin\frac{2\pi}{3}}{\cos\frac{2\pi}{3} + 2} = \frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2} + 2} = \frac{\sqrt{3}}{3},$$
$$\text{When } x = -\frac{2\pi}{9}, \quad y = \frac{\sin(3 \times -\frac{2\pi}{9})}{\cos(3 \times -\frac{2\pi}{9}) + 2} = \frac{\sin(-\frac{2\pi}{3})}{\cos(-\frac{2\pi}{3}) + 2} = \frac{-\frac{\sqrt{3}}{2}}{-\frac{1}{2} + 2} = -\frac{\sqrt{3}}{3}.$$

So $A(\frac{2\pi}{9}, \frac{\sqrt{3}}{3})$, $B(\frac{2\pi}{9}, -\frac{\sqrt{3}}{3})$ and $C(-\frac{2\pi}{9}, -\frac{\sqrt{3}}{3})$.

Method 1:

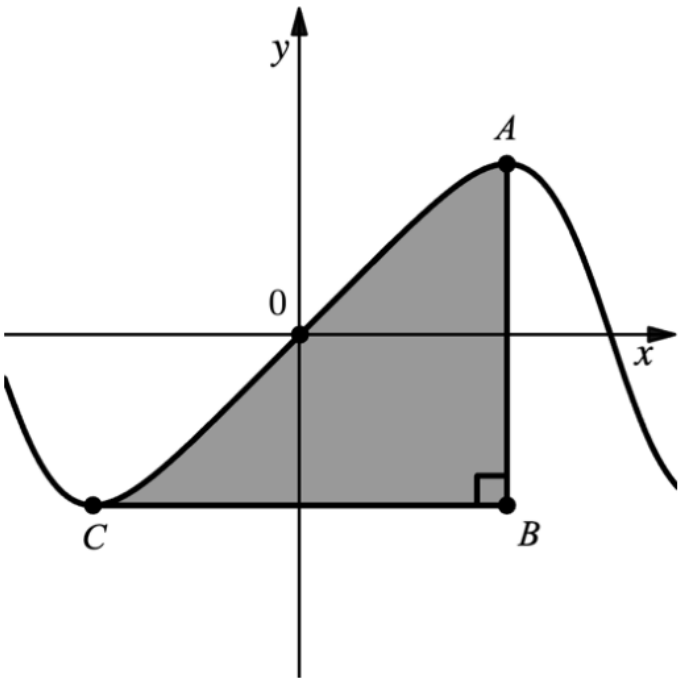
$$\int_{-2\pi/9}^{0} \frac{\sin 3x}{\cos 3x + 2} dx = \left[\sin 3x \times \frac{1}{-3\sin 3x} \times \ln|\cos 3x + 2|\right]_{-2\pi/9}^{0}$$
$$= \left[-\frac{1}{3}\ln|\cos 3x + 2|\right]_{-2\pi/9}^{0}$$
$$= -\frac{1}{3}\ln|\cos(3 \times 0) + 2| - \left(-\frac{1}{3}\ln|\cos(3 \times -\tfrac{2\pi}{9}) + 2|\right)$$
$$= -\frac{1}{3}\ln|\cos 0 + 2| - \left(-\frac{1}{3}\ln|\cos(-\tfrac{2\pi}{3}) + 2|\right)$$
$$= -\frac{1}{3}\ln|1 + 2| - \left(-\frac{1}{3}\ln|-\tfrac{1}{2} + 2|\right)$$
$$= -\frac{1}{3}\ln 3 - (-\tfrac{1}{3}\ln\tfrac{3}{2}) = \frac{1}{3}(\ln\tfrac{3}{2} - \ln 3)$$
$$= \frac{1}{3}\ln\frac{1}{2} = -\frac{1}{3}\ln 2.$$

$$\int_{0}^{2\pi/9} \frac{\sin 3x}{\cos 3x + 2} dx = \left[-\frac{1}{3}\ln|\cos 3x + 2|\right]_{0}^{2\pi/9}$$
$$= -\frac{1}{3}\ln|\cos(3 \times \tfrac{2\pi}{9}) + 2| - \left(-\frac{1}{3}\ln|\cos(3 \times 0) + 2|\right)$$
$$= -\frac{1}{3}\ln|\cos\tfrac{2\pi}{3} + 2| - \left(-\frac{1}{3}\ln|\cos 0 + 2|\right)$$

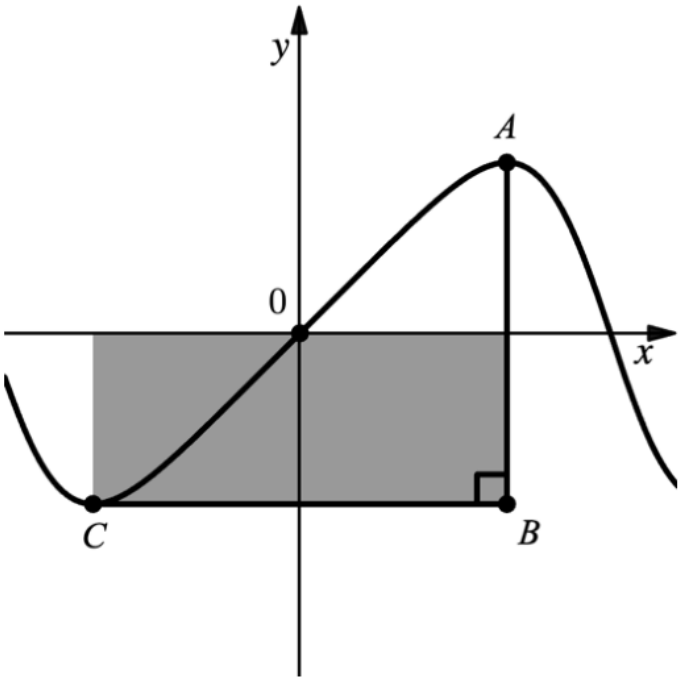
$$= -\frac{1}{3}\ln|-\tfrac{1}{2}+2| - \left(-\frac{1}{3}\ln|1+2|\right)$$
$$= -\frac{1}{3}\ln\frac{3}{2} - (-\tfrac{1}{3}\ln 3) = \frac{1}{3}(\ln 3 - \ln\tfrac{3}{2}) = \frac{1}{3}\ln 2.$$

$\therefore$ Area $= (\frac{2\pi}{9}\times\frac{\sqrt{3}}{3} - \frac{1}{3}\ln 2) + (\frac{1}{3}\ln 2) + (\frac{2\pi}{9}\times\frac{\sqrt{3}}{3}) = \frac{4\pi\sqrt{3}}{27}$.

Method 2: Use the symmetry of the graph.



is equivalent to:



$\therefore$ Area $= \frac{\sqrt{3}}{3}\times(\frac{2\pi}{9}+\frac{2\pi}{9}) = \frac{4\pi\sqrt{3}}{27}$.